WEDDERBURN'S THEOREM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; it is short but every step is a genuine reduction with its input named, and the closing sentence is the construction of the asserted isomorphism rather than a deferral. The statement opens with the author's own slogan environment, a comment-type environment of the chapter preamble, delivered as part of the counted statement.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The endomorphism ring of a simple submodule. Same chapter, Lemma lemma-simple-module. Supplies that the endomorphism ring of a simple submodule is a skew field.
- Rieffel's theorem. Same chapter, Lemma lemma-rieffel. Supplies that the algebra coincides with the bicommutant of a right ideal, which identifies the algebra with the endomorphisms of the chosen module over the skew field.

Source reference. Original lemma 0746: sources/stacks-brauer/pinned-original-source.tex, source lines 337–337. Navigation pointer invoked at proof line 369; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0745: sources/stacks-brauer/pinned-original-source.tex, source lines 309–309. Navigation pointer invoked at proof line 371; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:brauer.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Theorem 0.1. *Let A be a simple finite k-algebra. Then A is a matrix algebra over a finite k-algebra K which is a skew field.*

Proof. We may choose a simple submodule $M \subset A$ and then the k-algebra $K = \mathrm{End}_A(M)$ is a skew field, see Lemma 0746. By Lemma 0745 we see that $A = \mathrm{End}_K(M)$. Since K is a skew field and M is finitely generated (since $\dim_k(M) < \infty$) we see that M is finite free as a left K-module. It follows immediately that $A \cong \mathrm{Mat}(n \times n, K^{op})$. □